



October 2026

Assignment A3.1 – Three Worlds of 2041

The Creative Plan · How AI could reshape
our industry, our customers and our planet
over the next 15 years

Name : Mohammad Alkhatib

Section : ZMDT 2023 Fast Track 1 · Week 3



Course: [ZN-CSS-23F12M4S4] Zain 2023 Fast Track 1 – Creative Skills and Solutions

Prof. Name: Brendan Anglin

TAs: Lydia Mastori, Joanna Carrillo, Sophia Swain

Three Worlds of 2041

How AI could reshape our industry, our customers and our planet over the next 15 years

THREE SCENARIOS, NOT THREE PREDICTIONS

On 6 April 2026, four people watched Earth set behind the Moon. Fifteen years from now, the Moon may be a place where people go to work. This is not a dream. It is three futures we can already see coming, and the choices that decide which one we get.

IN THIS BRIEFING today's signals, our method, three worlds of 2041, the odds, and what to do from Monday.



Mohammad

Senior Futures Consultant, Em, Oli & Timmonds

Part 1 | The Creative Solution

Earth sets behind the Moon, photographed by the Artemis II crew, 6 April 2026. Image: NASA

The future is already leaking in

Four signals we can measure in 2026, found with two of the 4 lenses of innovation: harnessing trends and challenging orthodoxies. Every world grows out of them.

2026

2041



415 TWh

used by data centres in 2024, about 1.5% of world electricity, with about 945 TWh by 2030 as a base-case projection. (IEA)

Energy becomes AI's ceiling.



170M new jobs

created and 92M displaced by 2030 across major economic and technological trends, including AI; 39% of core skills change. (WEF)

Skills, not jobs, become the scarce asset.



54,000

debris objects larger than 10 cm circle Earth, plus about 1.2 million between 1 and 10 cm. (ESA)

Orbit is a finite resource we can pollute.



April 2026

Artemis II flies four astronauts around the Moon, the first crewed lunar flight since 1972. (NASA)

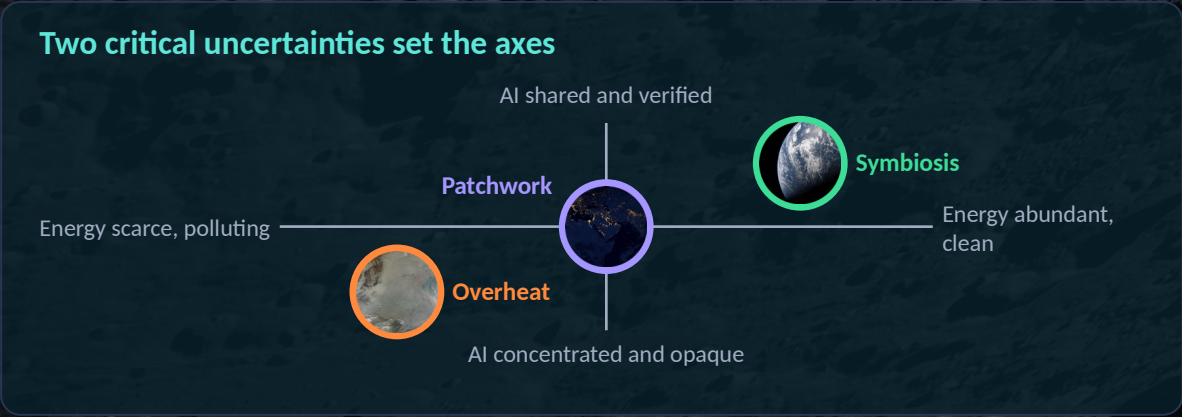
Space moves from exploration to economy.



Challenging orthodoxies. The old belief "AI is software" no longer holds. AI is an energy, materials and orbit industry, and that is exactly where our three worlds split apart.

The E,O&T Futures Engine

How the signals become worlds. Seven tools work like gears, and each produces a visible output on the slides shown. We first ran a controlled What if (keep AI access fixed, change only energy headroom), then combined both uncertainties into three worlds.



AI "bad copying"

We asked AI for dozens of variants of each world and kept the strange ones. Its mistakes became raw material.

Exaptation

Borrowed from air traffic control (one network, many vehicles) and multiplayer game design (rules that shape behaviour).

2041

Moonshot thinking: the upward What if
Shared, verified AI with abundant clean energy

Shared, verified AI with abundant clean energy



Robots on the Moon (NASA concept)


A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Layla, 34, grid engineer, Riyadh

Scene: 6 am. My AI agent has already sold my solar surplus and drafted my report.

Feel: free, and strangely idle.

Make sense: the machines now do the routine half of my job.

Worry: what am I still for?

Act: I join the council that designs our city's digital twin.

A future we want to be part of

The best case. We push the engine's upper branch to a moonshot, then live one day inside it through hard empathy.



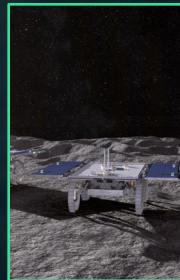
Mirror Cities

Customers try a flat or a lunar hotel in a live digital twin before buying. White-hat Octalysis: each purchase plants a real tree.



Life and transport

Electric pods and air taxis on one AI network, borrowed from air traffic control. Car parks become parks.



Space and the Moon

A crewed base at the lunar south pole mines ice for fuel. Robots build; the first tourists arrive.



Our industry

AI writes the analysis; consultants own the judgement. E,O&T move: launch a future-design practice.

5 Cs

Character

Layla, and every
client like her

Conflict

AI's hunger for energy

Cure

Clean power, efficient chips

Change

Abundance, shared
widely

Carry-out

Build the future we want

Still hard here: e-waste, attention overload, disputes over lunar mining.

Concentrated, opaque AI with scarce, polluting energy



Hardware in low orbit (NASA, ISS)



Omar, 41, logistics manager

Scene: in the depot, two delivery alerts contradict each other.

Feel: anxious; the rationing app also limits when I can charge the vans.

Make sense: one of the alerts may be synthetic.

Worry: I could send a driver onto an unsafe route.

Act: I verify both alerts through our data cooperative before dispatch.

Every utopia has a shadow

The honest shadow of World 1. Same engine, downward What if, told as a chain of cause and effect.

HOW IT HAPPENED, ONE "BECAUSE OF THAT" AT A TIME

1

AI demand outran clean power, so fossil plants kept running.

2

Heat and pollution rose,
and power was rationed.

3

A few players owned the models; deepfakes broke trust.

4

A debris cascade took out navigation and broadband satellites.



Customers

The metaverse is an escape; black-hat Octalysis (scarcity, loss) keeps people hooked.



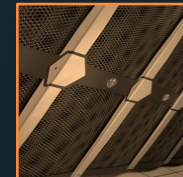
Life and transport

Empty robotaxis gridlock the cities.
The rich fly; everyone else waits.



Space and the Moon

Low orbit is a junkyard; the lunar base waits.



Our industry

Reports cost pennies. E,O&T move:
verification and resilience services.

5 Cs

Character
Omar

Conflict

AI outgrows clean power

Cure

None, in time

Change

Trust and orbit lost

Carry-out

Plan for the shadow first

PATCHWORK

2041

Tick-Tock: the middle path

Trust verified inside regional blocs, mixed energy between them

One planet, three rulebooks.



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Sara, 29, product designer, Stockholm

Scene: my AI agent is doing my weekly shopping.

Feel: relaxed, until it refuses my favourite app.

Make sense: that app has no trust seal in our bloc.

Worry: my own product could be locked out the same way.

Act: I design one product that passes all three blocs' trust rules.

Neither heaven nor collapse

The middle path, built by flipping two of today's facts with Tick-Tock. Here trust, not technology, decides who wins.

TICK People buy from companies.

TOCK AI agents buy from AI agents, so firms must design for machine customers.

TICK Data is free.

TOCK Data is owned and paid for, like energy.



Customers

Regional metaverses joined by one digital identity, so trust becomes the product.



Life and transport

Driverless in some regions, manual in others; the best cities blend both.



Space and the Moon

Shared on paper; blocs compete quietly for lunar ice and landing sites.



Our industry

Winners translate between blocs.
E,O&T move: a cross-bloc standards
advisory.

5 Cs

Character
Sara

Conflict

One world, many rulebooks

Cure

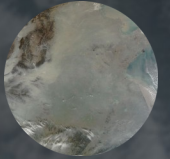
Shared trust seals

Change

Identity is the product

Carry-out

The future is a negotiation



One causal story inside one world

The worlds branch, but a story inside a world must follow cause and effect. All eight Story Spine steps for Omar in Overheat, a fictional rehearsal.



01 ONCE UPON A TIME

Omar ran a depot whose customers depended on daily deliveries.



02 EVERY DAY

His team dispatched the vans on the AI's recommendation.



03 BUT ONE DAY

Two alerts conflicted, and the planned van could not charge.



04 BECAUSE OF THAT

Omar paused the recommendation before sending a driver.



05 BECAUSE OF THAT

He checked the records and called the depot to verify the route.



06 BECAUSE OF THAT

He used the manual fallback and logged why the decision changed.



07 UNTIL FINALLY

A verified alternative kept the delivery moving.



08 AND EVER SINCE THEN

The team reviews every exception with a Decision Passport.

Takeaway (5 Cs carry-out). The cure is not perfect AI. It is a process that keeps evidence, ownership and a fallback visible.

Prove it is useful before scaling it

One client, one workflow, one limited rehearsal. All activities and gates below are proposed; no client commitment or benefit is claimed.

DAYS 1 TO 30



Define and observe

OWNERS Client owner and consultant

Map one workflow, record a manual baseline and interview five users.

GATE: agree success, error severity, data permissions and limits.

DAYS 31 TO 60



Prototype and rehearse

OWNERS Technical lead and client users

Build a simulation and a Decision Passport; rehearse all three worlds.

GATE: five users can name the evidence, owner and fallback.

DAYS 61 TO 90



Stress-test and decide

OWNERS Sponsor and independent reviewer

Run ten scripted cases: low power, false input, provider outage and normal operation.

GATE: every critical fallback works; no unresolved material error.

Continue, change or stop. Scale only when the pilot shows client-valued benefit and acceptable resource use.

Pilot limit. Five users and ten cases support learning, not proof of general reliability.

What I am taking with me

The tools that resonated, the change in my thinking, and why it matters in my professional life.

Three ideas from this course changed how I think about planning: possibility trees, working backwards, and counterfactual thinking. Together they gave me a structured language for something my work has always demanded: decisions about a future nobody can see.

In twenty years of network engineering and digital transformation, I have learned that a platform chosen today may still shape a business in ten years. Building the three worlds of 2041 for this assignment showed me a different way to approach that kind of decision. Drawing the possibility tree forced me to give each branch a probability, and the less likely branches raised questions a single forecast would never have asked. Working backwards from a 2041 press release, the way Amazon writes the press release before the product exists, made the first steps for 2026 obvious. Counterfactual thinking kept the exercise honest: writing Overheat next to Symbiosis stopped me from falling in love with the optimistic future.

What resonated most is that creativity here is not a talent but a structured skill, wild ideas plus strategy.



20 years

in networks and digital transformation,
where decisions have long consequences

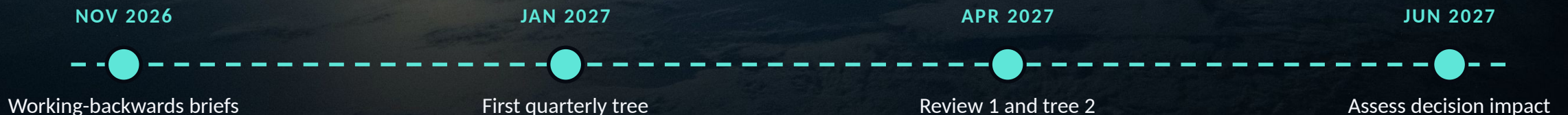
How I will put it into practice

A realistic timetable, a review loop and a personal routine that continues beyond this course.

My plan is concrete. From November, every new technical proposal I write will open with a one-page working backwards brief describing the result for the customer in three years, the assumptions behind it and the first test. Once a quarter, starting in January 2027, I will build a possibility tree for one roadmap decision, with three branches, nine sub-branches and probabilities, and review it three months later to see which branch reality followed. Before each major design review I will run one upward and one downward counterfactual with the team and record the risks it surfaces.

Personally, I live between Riyadh and Stockholm, and I will turn those flights into fuzzy focus time: no screens, just my stimuli queue and a commonplace book, so slow hunches have somewhere to land.

I will measure success simply: by June 2027, I will have built two possibility trees, completed the first three-month review, and recorded whether the findings changed a roadmap decision.



Sources

European Space Agency. (2025). ESA space environment report 2025. https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025

International Energy Agency. (2025). Energy and AI. <https://www.iea.org/reports/energy-and-ai>

NASA. (2026). Artemis II mission. <https://www.nasa.gov/mission/artemis-ii/>

World Economic Forum. (2025). The future of jobs report 2025. <https://www.weforum.org/publications/the-future-of-jobs-report-2025/>

UNEP. (2022). Sand and sustainability: 10 strategic recommendations to avert a crisis. United Nations Environment Programme.

Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Making AI less "thirsty". arXiv:2304.03271.

NASA. (2023). NASA achieves water recovery milestone on International Space Station.

NASA. (2021). Astronauts harvest first chile peppers grown on the space station (Plant Habitat-04).

ESA. (2020). ESA opens oxygen plant: making air out of moondust.

Food On Demand. (2026, January 21). Zipline flies past 2 million deliveries.

Business Standard. (2025, March 19). What astronauts eat on the ISS (NASA food stock of 3.8 lb per astronaut per day).

Adams, K. (2007). How to improvise a full-length play. Allworth Press. (Story Spine)

Chou, Y. (2015). Actionable gamification: Beyond points, badges, and leaderboards. Octalysis Media.

Flyvbjerg, B., & Gardner, D. (2023). How big things get done. Currency.

Gibson, R. (2015). The four lenses of innovation. Wiley.

Henry, T. (2011). The accidental creative. Portfolio.

Johnson, S. (2010). Where good ideas come from. Riverhead.

Lee, K.-F., & Chen, Q. (2021). AI 2041: Ten visions for our future. Currency.

McGonigal, J. (2022). Imaginable. Spiegel & Grau.

Image credits

Slide 1: "A Breathtaking Earthset from Orion", Artemis II crew, 6 April 2026. NASA (art002e021278).

Slide 3: Night Earth observation from the ISS, Expedition 40. NASA (iss040e085126).

Slides 2, 4 and 14: "Home in the Distance", Artemis II. NASA (art002e020922).

Slide 5: "Earth in View", Artemis II (art002e023506); CADRE rovers concept, NASA/JPL-Caltech (PIA26296).

Slide 6: "Haze over eastern China", NASA Goddard (GSFC_20171208_Archive_e001269); ISS Expedition 35 (iss035e017673).

Slides 7, 10 and 17: "Black Marble: Africa, Europe, and the Middle East", NASA Goddard (GSFC_20171208_Archive_e001591).

Slides 5 to 7 and 16: NASA VR training (jsc2026e000427), Advanced Air Mobility concepts (AAM-Concept-2021, automation concept), Mission Control (jsc2026e019925), Pleiades supercomputer (ARC-2008-ACD08-0272-005).

Slide 8: NASA drone flight test (AFRC2026-0152-22); Cygnus in Canadarm2 (iss071e418235); Expedition 66 chile peppers (iss066e023179).

Slide 9: ISS water recovery (iss072e143163); sand dunes and irrigation in Saudi Arabia (PIA11069); Earth limb and Moon (iss051e039977); Veg-01 harvest (iss040e009126); JWST golden mirror (GSFC_20171208_Archive_e000368).

World thumbnails on slides 2, 4, 10, 12, 13 and 17 reuse the images of slides 5, 6 and 7.

All NASA images via the NASA Image and Video Library, <https://images.nasa.gov> (public domain).

Portrait: own photograph. Icons: Font Awesome Free (CC BY 4.0), Material Design Icons (Apache 2.0), Game-icons.net (CC BY 3.0).

Use of AI

I acknowledge the use of Claude (<https://claude.ai>), Canva AI (<https://www.canva.com>), Superdesign (<https://superdesign.dev>) and Miro (<https://miro.com>) to research and verify sources, structure and draft sections, generate the mood board and app visuals, and arrange and format the document. The output was used to support my own work .